

1. What is a CI/CD Pipeline?

CI/CD = Continuous Integration / Continuous Deployment (or Delivery)

It is an **automated process** to build, test, and release software quickly and safely.

Think of it like this:

Developer writes code → code is tested automatically → app is deployed automatically

CI (Continuous Integration)

When developers write code and push it to a tool like [GitHub](#) or [GitLab](#):

- Code gets **merged** into a shared repository
- Automated **build** happens
- Automated **tests** run
- If tests pass → move to next step

Example tools:

- [Jenkins](#)
- [GitHub Actions](#)
- [GitLab CI/CD](#)

CD (Continuous Deployment/Delivery)

After testing, the application is automatically moved to servers/environments.

- **Continuous Delivery** → ready for release, but someone clicks “Deploy”
- **Continuous Deployment** → automatically goes live without manual approval

2. How does an application/product go live?

Typical flow:

Developer writes code

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Push code to Git (GitHub)

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CI pipeline runs

(Build + Unit Test + Code Check)

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Deploy to Dev Environment

↓

QA/Testing Team tests

↓

Deploy to Staging/UAT

(Client or business verifies)

↓

Approval given

↓

Deploy to Production

↓

Users can access the live app

Example: If you update **Amazon payment screen**:

1. Developer changes payment code
2. Pushes code to Git
3. CI pipeline checks if code is correct
4. Deploys to test server
5. QA tests payment works
6. Business approves
7. Deploy to **Production (Live)**
8. Customers now use it

3. Different environments in a project

Local Environment

Your own laptop/system.

- Developer writes and tests code here
- Not shared with users

Example: Running your Spring Boot app on localhost

Development (Dev) Environment

Shared environment for developers.

- Team integrates code
 - Basic testing
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QA / Testing Environment

QA = Quality Assurance

- Testers check bugs
 - Functional testing happens
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UAT (User Acceptance Testing) / Staging

Very close to real production.

- Client/business checks if everything is correct
 - Final approval happens here
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Production (Prod)

The live environment.

- Real users access the application
- Must be stable

Example:

[Amazon India](#) live website = Production

Simple environment flow

Local → Dev → QA/Test → UAT/Staging → Production